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Chapter 5

1. A single CPU can many prosses at some time because A single CPU uses a technique called Time sharing.
   - CPU gives a small amount of time to each process.
   - After completing its time, CPU switches to another process
   - This switching happens very quickly

2. The opp operating system gives every process its own separate memory space
   This is called memory isolation
   Reasons:
   1. Security
   2. protection → If chrome crashes, it normally won't crash us cafe
   3. Privacy

3. The opperating System manages memory using a technique called virtual memory when the process starts, the OS provides it with its own virtual adress place. The OS & hardware (MMU) translate virtual addresses into physical memory addresses using page tables
   - Process memory stores
     Code, Data, Heap, Stack

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4.(a) Memory Management

1. OS: Allocates memory to different processes so they don't interfere with each other
2. College ops Team: Assigns classrooms to different classes so two classes don't get the same room
3. If removed: Processe may interfere with each other. In college, two classes may arrive in the same classroom, creating confusion

b) Process Management

1. OS: manages multiple processes and gives CPU time to each process
2. OPS → Manage class schedules & timings so different activities run smoothly
3. If removed: In OS, processes may not get proper CPU time. In college classes & activities may overlap

c) I/O - Device Management

1. OS: Manage input & output devices like keyboard, printers & speakers
2. College OPS Teame: maintain projectors, ACs, electricity & other classroom equipment

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3. if removed: in OS, devices may not work properly with applications, In college lectures may be interrupted because equipment is not working